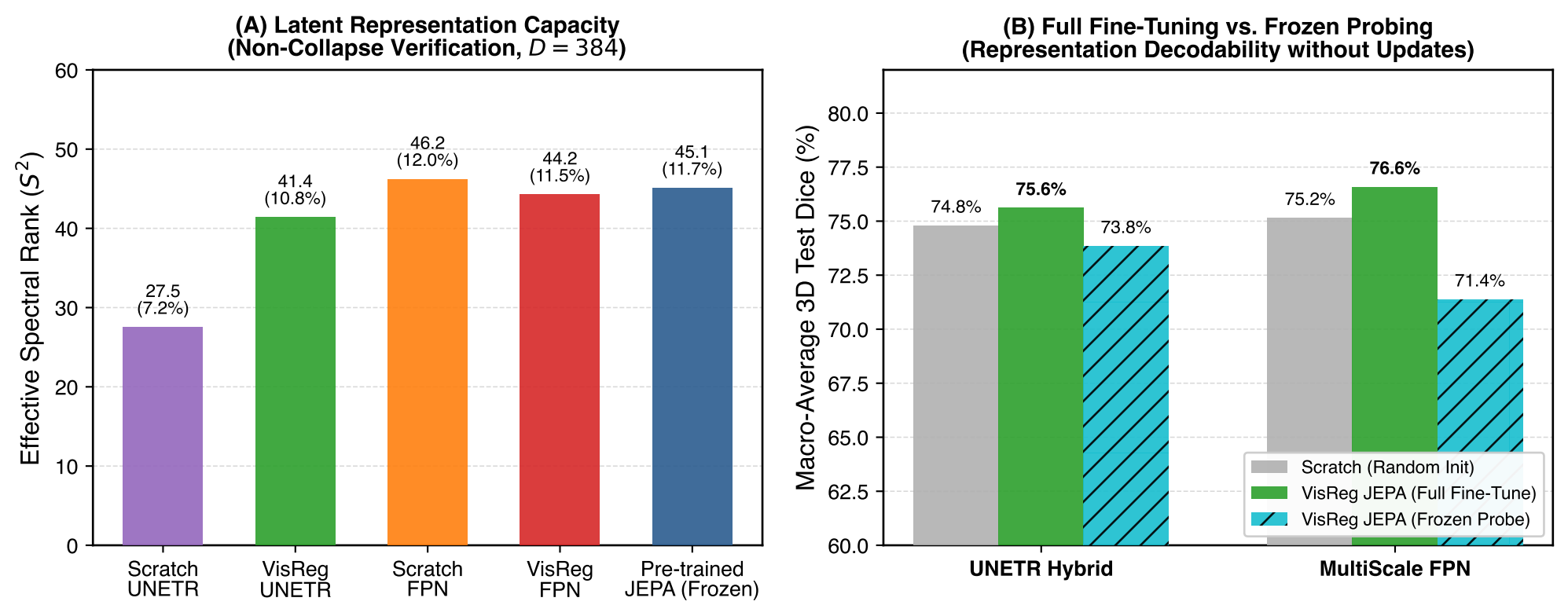
(A) Latent Representation Capacity
(Non-Collapse Verification, $D = 384$)
Effective Spectral Rank (S^2)
60
50
40
30
20
10
0
27.5
(7.2%)
41.4
(10.8%)
46.2
(12.0%)
44.2
(11.5%)
45.1
(11.7%)
Scratch
UNETR
VisReg
UNETR
Scratch
FPN
VisReg
FPN
Pre-trained
JEPA (Frozen)
(B) Full Fine-Tuning vs. Frozen Probing
(Representation Decodability without Updates)
Macro-Average 3D Test Dice (%)
80.0
77.5
75.0
72.5
70.0
67.5
65.0
62.5
60.0
74.8%
75.6%
73.8%
75.2%
76.6%
71.4%
UNETR Hybrid
MultiScale FPN
Scratch (Random Init)
VisReg JEPA (Full Fine-Tune)
VisReg JEPA (Frozen Probe)